

1. Create a Table with Foreign Key Constraint: Write a query to create a table named ENROLLMENTS with the following columns: EnrollmentID (NUMBER, primary key), StudentID (NUMBER), CourseID (NUMBER), and a foreign key constraint referencing the StudentID column from ENROLLED_STUDENTS and the CourseID column from COURSES.

```
CREATE TABLE ENROLLMENTS (  
    EnrollmentID NUMBER PRIMARY KEY,  
    StudentID NUMBER,  
    CourseID NUMBER,  
  
    CONSTRAINT FK_STUDENT  
    FOREIGN KEY (StudentID)  
    REFERENCES ENROLLED_STUDENTS(StudentID),  
  
    CONSTRAINT FK_COURSE  
    FOREIGN KEY (CourseID)  
    REFERENCES COURSES(CourseID)  
);
```

2. Using the tables previously given (ENROLLED_STUDENTS, COURSES, ENROLLMENTS): Insert a Record into ENROLLED_STUDENTS: Write a query to insert a new student with the following details: StudentID = 101, FirstName = 'John', LastName = 'Doe', BirthDate = '1999-12-01', and PhoneNumber = '123-456-7890'.

```
INSERT INTO ENROLLED_STUDENTS  
(StudentID, FirstName, LastName, BirthDate, PhoneNumber)  
VALUES  
(101, 'John', 'Doe', TO_DATE('1999-12-01','YYYY-MM-DD'), '123-456-7890');
```

3. Update a Record in ENROLLED_STUDENTS: Write a query to update the PhoneNumber of the student with StudentID = 101 to '987-654-3210'.

UPDATE ENROLLED_STUDENTS

SET PhoneNumber = '987-654-3210'

WHERE StudentID = 101;

4. Delete a Record from ENROLLED_STUDENTS: Write a query to delete the student with StudentID = 101 from the ENROLLED_STUDENTS table.

DELETE FROM ENROLLED_STUDENTS

WHERE StudentID = 101;

5. Write a query to insert the following courses into the COURSES table: CourseID = 201, CourseName = 'Database Systems', Credits = 3 CourseID = 202, CourseName = 'Operating Systems', Credits = 4

INSERT INTO COURSES

(CourseID, CourseName, Credits)

VALUES

(201, 'Database Systems', 3);

INSERT INTO COURSES

(CourseID, CourseName, Credits)

VALUES

(202, 'Operating Systems', 4);

6. Insert Records into ENROLLMENTS with Foreign Keys: Write a query to enroll the student with StudentID = 102 in the course CourseID = 201 and another query to enroll the same student in the course CourseID = 202.

INSERT INTO ENROLLMENTS

```
(EnrollmentID, StudentID, CourseID)
VALUES
(301, 102, 201);
```

```
INSERT INTO ENROLLMENTS
(EnrollmentID, StudentID, CourseID)
VALUES
(302, 102, 202);
```

7. Write a query to update the CourseName of CourseID = 201 to 'Advanced Database Systems'.

```
UPDATE COURSES
SET CourseName = 'Advanced Database Systems'
WHERE CourseID = 201;
```

8. Write a query to delete the course with CourseID = 202 from the COURSES table.

```
DELETE FROM COURSES
WHERE CourseID = 202;
```

9. Write a query to insert a new enrollment for StudentID = 103 in CourseID = 201, but only if that student isn't already enrolled in the course.

```
INSERT INTO ENROLLMENTS (EnrollmentID, StudentID, CourseID)
SELECT 303, 103, 201
FROM DUAL
WHERE NOT EXISTS (
    SELECT *
    FROM ENROLLMENTS
    WHERE StudentID = 103
    AND CourseID = 201
);
```

10. Write a query to retrieve the details of all students enrolled in the course with CourseID = 201.

```
SELECT s.StudentID,  
       s.FirstName,  
       s.LastName,  
       s.BirthDate,  
       s.PhoneNumber  
FROM ENROLLED_STUDENTS s  
JOIN ENROLLMENTS e  
ON s.StudentID = e.StudentID  
WHERE e.CourseID = 201;
```

11. Write a query to retrieve all the courses from the COURSES table that have more than 3 credits.

```
SELECT *  
FROM COURSES  
WHERE Credits > 3;
```

12. Write a query to truncate all the data in the ENROLLMENTS table without deleting the table itself.

```
TRUNCATE TABLE ENROLLMENTS;
```